

Modal Invariants and a Sharp Stability Threshold for Quadratic Symplectic Euler

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Abstract

Katona, Wang, and Wibisono give a closed-form modified Hamiltonian for symplectic Euler applied to a separable quadratic Hamiltonian. This note recasts that formula mode by mode in the positive-definite case. A linear canonical transformation produces independent two-dimensional modes, each with an explicit quadratic invariant, while the modified Hamiltonian is a frequency-dependent reweighting of those invariants. The same decomposition shows that

$$\eta^2 \rho(BC) < 1$$

is both the exact convergence condition for the modified-Hamiltonian series and the exact power-boundedness threshold of the discrete dynamics. It also identifies spectral anisotropy, rather than dimension alone, as the source of the difference between the modified and elementary invariants.

1 Modal reduction

Fix a stepsize $\eta > 0$ and consider

$$H(p, q) = p^\top Bp + q^\top Cq, \quad B, C \in \mathbb{R}^{d \times d}, \quad B, C \succ 0, \quad (1)$$

and the symplectic Euler convention that updates p first:

$$p^+ = p - 2\eta Cq, \quad q^+ = q + 2\eta Bp^+. \quad (2)$$

This is the quadratic instance of the Hamiltonian representation of alternating mirror descent studied in [2, 1].

Diagonalize the symmetric positive-definite matrix

$$C^{1/2}BC^{1/2} = U\Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d),$$

with U orthogonal and $\lambda_i > 0$. Set

$$D := \Lambda^{1/2} = \text{diag}(d_1, \dots, d_d), \quad d_i := \sqrt{\lambda_i}, \quad S := C^{-1/2}U\Lambda^{1/4}.$$

Since $BC = C^{-1/2}(C^{1/2}BC^{1/2})C^{1/2}$, the λ_i are also the eigenvalues of BC , and $\rho(BC) = \max_i \lambda_i$.

Theorem 1 (Modal invariants and the exact threshold). *Define the linear change of variables*

$$u = S^\top p, \quad v = S^{-1}q.$$

Then:

1. *The change is canonical, and both the Hamiltonian and the update separate into scalar modes:*

$$H = \sum_{i=1}^d d_i(u_i^2 + v_i^2), \quad u_i^+ = u_i - 2\eta d_i v_i, \quad v_i^+ = v_i + 2\eta d_i u_i^+. \quad (3)$$

2. Writing $x_i := \eta d_i$, the i -th mode preserves

$$I_i := d_i(u_i^2 + v_i^2 - 2x_i u_i v_i). \quad (4)$$

If $\eta^2 \rho(BC) < 1$, the modified Hamiltonian in [1, Theorem 12 and Appendix C] is

$$\tilde{H}_\eta = \sum_{i=1}^d T_\eta(\lambda_i) I_i, \quad T_\eta(\lambda) := \frac{\arcsin(\eta\sqrt{\lambda})}{\eta\sqrt{\lambda}\sqrt{1-\eta^2\lambda}}. \quad (5)$$

3. The following are equivalent:

$$\eta^2 \rho(BC) < 1;$$

(2) is power-bounded;

$$\sum_i I_i = p^\top B p + q^\top C q - 2\eta p^\top B C q \text{ is positive definite;}$$

the modified-Hamiltonian power series converges absolutely as a quadratic-form-valued series.

At equality, a highest-frequency mode has a nontrivial Jordan block at -1 ; beyond it, at least one mode is hyperbolic.

Proof. The block-diagonal map $\text{diag}(S^\top, S^{-1})$ preserves the standard symplectic form. Moreover,

$$S^{-1} B S^{-\top} = S^\top C S = D,$$

which gives (3). For one mode,

$$\begin{pmatrix} u_i^+ \\ v_i^+ \end{pmatrix} = M_i \begin{pmatrix} u_i \\ v_i \end{pmatrix}, \quad M_i = \begin{pmatrix} 1 & -2x_i \\ 2x_i & 1 - 4x_i^2 \end{pmatrix}.$$

With $G_i = \begin{pmatrix} 1 & -x_i \\ -x_i & 1 \end{pmatrix}$, direct multiplication gives $M_i^\top G_i M_i = G_i$, proving (4). Also $S D^2 S^{-1} = BC$, so summing the modal invariants gives the displayed expression in the original variables. Since distinct modes have vanishing mutual Poisson brackets, the scalar closed form from [1, Appendix C] adds across modes and yields (5).

Finally,

$$\det M_i = 1, \quad \text{tr } M_i = 2 - 4x_i^2.$$

Thus M_i has distinct conjugate eigenvalues on the unit circle exactly when $0 < x_i < 1$, a nontrivial Jordan block at $x_i = 1$, and a real reciprocal pair with one eigenvalue outside the unit circle when $x_i > 1$. This proves the power-boundedness claim. The eigenvalues of G_i are $1 \pm x_i$, giving the positive-definiteness claim. Finally, $T_\eta(\lambda_i)$, as a power series in η , has its nearest singularities at $\eta = \pm 1/\sqrt{\lambda_i}$. No cancellation can enlarge this radius, since $d_i T_\eta(\lambda_i)$ is the coefficient of u_i^2 in the i -th modal block. Hence the full quadratic series has radius

$$\min_i \frac{1}{\sqrt{\lambda_i}} = \frac{1}{\sqrt{\rho(BC)}}.$$

□

¹The η -weighted cross term used here agrees with Lemma 21 and Equation (80) of [1]; it is also directly preserved by (3).

2 What the modal weights reveal

Let $\mathcal{S}_\eta := \sum_i I_i$ denote the elementary quadratic invariant. The function

$$x \mapsto \frac{\arcsin x}{x\sqrt{1-x^2}}$$

is strictly increasing on $0 < x < 1$, as follows directly from its power series in x^2 , whose nonconstant coefficients are positive.

Corollary 2 (Spectral anisotropy). *Assume that $\eta^2 \rho(BC) < 1$. If $\lambda_1 = \dots = \lambda_d = \lambda$, then $\tilde{H}_\eta = T_\eta(\lambda)\mathcal{S}_\eta$. If BC has at least two distinct eigenvalues, then $\tilde{H}_\eta$ and $\mathcal{S}_\eta$ are functionally independent on the open dense set of points with nonzero components in two eigenspaces carrying distinct eigenvalues.*

Proof. The first claim follows from (5). For the second, if $\nabla \tilde{H}_\eta = \alpha \nabla \mathcal{S}_\eta$, then on every active modal component the invertibility of G_i forces $T_\eta(\lambda_i) = \alpha$. Strict monotonicity makes this impossible for two active modes with distinct eigenvalues. $\square$

Thus higher dimension alone does not make $\tilde{H}_\eta$ functionally independent of $\mathcal{S}_\eta$: for an isotropic spectrum, the two remain proportional. Generic independence already appears once two distinct modal frequencies are present. The modal reduction makes the eigenvalue-wise condition underlying [1, Appendix C] explicit: in the positive-definite case, $\eta^2 \rho(BC) < 1$ is the exact threshold for both absolute convergence and power-boundedness, whereas $\eta^2 \|BC\|_2 < 1$ is a convenient sufficient condition. The latter can be strict when BC is nonnormal. This concerns the qualitative threshold; the size of the power bound may still depend on the conditioning of the canonical transformation.

3 Beyond quadratics

For $H(p, q) = F(p) + G(q)$ with $F, G \in C^2$, define

$$B(p) = \frac{1}{2} \nabla^2 F(p), \quad C(q) = \frac{1}{2} \nabla^2 G(q).$$

Assume throughout this section that these Hessian blocks are positive definite at the points under consideration. The linearization of the continuous Hamiltonian vector field at (p, q) has eigenvalues

$$\pm 2i \sqrt{\lambda_i (C(q)^{1/2} B(p) C(q)^{1/2})},$$

so the corresponding local modal frequencies are

$$2 \sqrt{\lambda_i (C(q)^{1/2} B(p) C(q)^{1/2})}.$$

For the symplectic Euler map itself, however, the one-step Jacobian has the same block form as the quadratic update (2), with $B = B(p^+)$ and $C = C(q)$. Under the positivity assumption, each step therefore admits a local modal reduction, but the modal frequencies and canonical frame may change from step to step. In particular, one-step ellipticity alone need not imply power-bounded products.

This suggests a more specific question: under a uniform gap from the local threshold, Hessian-Lipschitz assumptions, and suitable spectral-gap control, can one bound the coupling induced by the changing canonical frame and obtain an approximate modal invariant or a trajectory-dependent sufficient condition for a fixed stepsize? In the AMD setting, can the mirror maps be chosen to keep the local spectrum well-conditioned and the frame drift small?

References

- [1] J. E. Katona, X. Wang, and A. Wibisono. A symplectic analysis of alternating mirror descent. *Journal of Machine Learning Research*, 27(44):1–61, 2026.
- [2] A. Wibisono, M. Tao, and G. Piliouras. Alternating mirror descent for constrained min–max games. In *Advances in Neural Information Processing Systems*, volume 35, 2022.